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Information Systems
Labwork 3

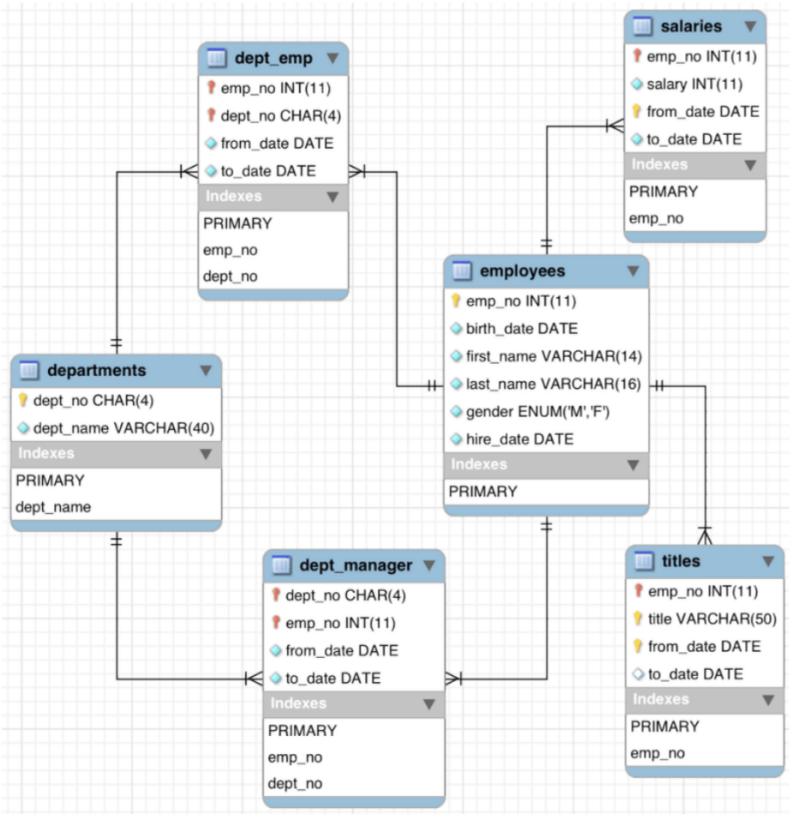


Figure 1: `employees` database schema

Database queries

Write SQL queries to get the following information:

- All info of all employees (output shortened)

```sql
USE employees;
SELECT * FROM employees
ORDER BY emp_no;
```

Output:

```
+--------+------------+------------+--------------+--------+------------+
| emp_no | birth_date | first_name | last_name    | gender | hire_date  |
+--------+------------+------------+--------------+--------+------------+
|  10001 | 1953-09-02 | Georgi     | Facello      | M      | 1986-06-26 |
|  10002 | 1964-06-02 | Bezalel    | Simmel       | F      | 1985-11-21 |
|  10003 | 1959-12-03 | Parto      | Bamford      | M      | 1986-08-28 |
```

```
|  10004 | 1954-05-01 | Chirstian   | Koblick      | M      | 1986-12-01 |
|  10005 | 1955-01-21 | Kyoichi     | Maliniak     | M      | 1989-09-12 |
|  10006 | 1953-04-20 | Anneke      | Preusig      | F      | 1989-06-02 |
+--------+------------+-------------+--------------+--------+------------+
```

- All info of all departments

```
USE employees;
SELECT * FROM departments;
```

Output:

```
+---------+--------------------+
| dept_no | dept_name          |
+---------+--------------------+
| d009    | Customer Service   |
| d005    | Development        |
| d002    | Finance            |
| d003    | Human Resources    |
| d001    | Marketing          |
| d004    | Production         |
| d006    | Quality Management |
| d008    | Research           |
| d007    | Sales              |
+---------+--------------------+
```

- Full name of all employees (output shortened)

```
USE employees;
SELECT
    emp_no,
    CONCAT(first_name, ' ', last_name) AS full_name
FROM employees
ORDER BY emp_no;
```

Output:

```
+--------+-----------------------+
| emp_no | full_name             |
+--------+-----------------------+
|  10001 | Georgi Facello        |
|  10002 | Bezalel Simmel        |
|  10003 | Parto Bamford         |
|  10004 | Chirstian Koblick     |
|  10005 | Kyoichi Maliniak      |
|  10006 | Anneke Preusig        |
+--------+-----------------------+
```

- Names of all departments

```
USE employees;
SELECT dept_name FROM departments;
```

Output:

```
+--------------------+
| dept_name          |
+--------------------+
| Customer Service   |
| Development        |
| Finance            |
| Human Resources    |
| Marketing          |
| Production         |
| Quality Management |
```

```
| Research           |
| Sales              |
+--------------------+
```

- Full names of employees working in 'Sales' department (output shortened)

```sql
SELECT
    e.emp_no AS emp_no_sales,
    CONCAT(e.first_name, ' ', e.last_name) AS full_name
FROM employees AS e
INNER JOIN dept_emp AS de ON e.emp_no = de.emp_no
INNER JOIN departments AS d ON de.dept_no = d.dept_no
WHERE d.dept_name = "Sales";
```

Output:

```
+--------------+-------------------------------+
| emp_no_sales | full_name                     |
+--------------+-------------------------------+
|        10002 | Bezalel Simmel                |
|        10016 | Kazuhito Cappelletti          |
|        10034 | Bader Swan                    |
|        10041 | Uri Lenart                    |
|        10050 | Yinghua Dredge                |
|        10053 | Sanjiv Zschoche               |
|        10060 | Breannda Billingsley          |
+--------------+-------------------------------+
```

- Full names of male employees working in 'Finance' department (output shortened)

```sql
SELECT
    e.emp_no AS emp_no_sales,
    CONCAT(e.first_name, ' ', e.last_name) AS full_name
FROM employees AS e
INNER JOIN dept_emp AS de ON e.emp_no = de.emp_no
INNER JOIN departments AS d ON de.dept_no = d.dept_no
WHERE d.dept_name = "Finance" AND e.gender = "M";
```

Output:

```
+--------------+-------------------------------+
| emp_no_sales | full_name                     |
+--------------+-------------------------------+
|        10050 | Yinghua Dredge                |
|        10080 | Premal Baek                   |
|        10132 | Ayakannu Skrikant             |
|        10144 | Marla Brendel                 |
|        10146 | Chenyi Syang                  |
|        10147 | Kazuhito Encarnacion          |
|        10165 | Miyeon Macedo                 |
|        10173 | Shrikanth Mahmud              |
|        10177 | Pragnesh Iisaka               |
+--------------+-------------------------------+
```

- Salaries of female employees working in 'Marketing' department (output shortened)

```sql
SELECT
    e.emp_no,
    s.salary,
    CONCAT(e.first_name, ' ', e.last_name) AS full_name
FROM employees AS e
INNER JOIN dept_emp AS de ON e.emp_no = de.emp_no
INNER JOIN departments AS d ON de.dept_no = d.dept_no
INNER JOIN salaries AS s ON de.emp_no = s.emp_no
WHERE d.dept_name = 'Marketing' AND e.gender = 'F';
```

Output:

```
+--------+--------+-----------------------------+
| emp_no | salary | full_name                   |
+--------+--------+-----------------------------+
|  10017 |  71380 | Cristinel Bouloucos         |
|  10017 |  75538 | Cristinel Bouloucos         |
|  10017 |  79510 | Cristinel Bouloucos         |
|  10017 |  82163 | Cristinel Bouloucos         |
|  10017 |  86157 | Cristinel Bouloucos         |
|  10017 |  89619 | Cristinel Bouloucos         |
|  10017 |  91985 | Cristinel Bouloucos         |
|  10017 |  96122 | Cristinel Bouloucos         |
|  10017 |  98522 | Cristinel Bouloucos         |
|  10017 |  99651 | Cristinel Bouloucos         |
|  10140 |  61352 | Yucel Auria                 |
|  10140 |  61609 | Yucel Auria                 |
+--------+--------+-----------------------------+
```

- Full names of employees who have the same last name as their manager (output shortened)

```
SELECT CONCAT(e.first_name, ' ', e.last_name) AS full_name
FROM employees AS e
INNER JOIN dept_emp AS de ON e.emp_no = de.emp_no
INNER JOIN dept_manager AS dm ON de.dept_no = dm.dept_no
INNER JOIN employees AS m ON dm.emp_no = m.emp_no
WHERE e.last_name = m.last_name AND e.emp_no <> m.emp_no;
```

Output:

```
+-------------------------+
| full_name               |
+-------------------------+
| Herbert Markovitch      |
| Zita Markovitch         |
| Khun Markovitch         |
| Nalini Markovitch       |
| Sibyl Markovitch        |
| JoAnne Markovitch       |
| Weiru Markovitch        |
| Kerhong Markovitch      |
| Baocai Markovitch       |
+-------------------------+
```

- Full names of managers who have been doing the job at least twice

```
SELECT CONCAT(e.first_name, ' ', e.last_name) AS full_name
FROM employees AS e
INNER JOIN dept_manager AS dm ON e.emp_no = dm.emp_no
GROUP BY dm.emp_no, e.first_name, e.last_name
HAVING COUNT(*) >= 2;
```

Output:

```
+-----------------------------+
```

- Full names of employees who was paid more than $100000

```
SELECT CONCAT(e.first_name, ' ', e.last_name) AS full_name
FROM employees AS e
INNER JOIN dept_emp AS de ON e.emp_no = de.emp_no
INNER JOIN salaries AS s ON e.emp_no = s.emp_no
WHERE s.salary > 100000
GROUP BY e.emp_no;
```

Output:

```
+----------------------------+
| full_name                  |
+----------------------------+
| Kwee Schusler              |
| Charlene Brattka           |
| Xinglin Eugenio            |
| Dung Baca                  |
| Zissis Pintelas            |
| Zhenbing Perng             |
| Itzchak Lichtner           |
| Debatosh Khasidashvili     |
| Shrikanth Mahmud           |
| Duro Sidhu                 |
| Marko Auria                |
+----------------------------+
```

- Names of all departments that have employees paid more than $100000

```sql
SELECT d.dept_name
FROM dept_emp AS de
INNER JOIN departments AS d ON de.dept_no = d.dept_no
INNER JOIN salaries AS s ON de.emp_no = s.emp_no
WHERE s.salary > 100000
GROUP BY d.dept_name;
```

Output:

```
+--------------------+
| dept_name          |
+--------------------+
| Customer Service   |
| Development        |
| Finance            |
| Human Resources    |
| Marketing          |
| Production         |
| Quality Management |
| Research           |
| Sales              |
+--------------------+
```

Merged departments

Due to changes in company's organization, the following departments are merged:

- Development (`d005`) and Research (`d008`) → Research and Development (`d010`)
- Marketing (`d001`) and Sales (`d007`) → Marketing and Sales (`d011`)

Necessary SQL queries to implement these changes to the employees database:

- Create new departments: `d005`, `d008` → `d010` and `d001`, `d007` → `d011`

```sql
INSERT INTO departments (dept_no, dept_name) VALUES
('d010', 'Research and Development'), ('d011', 'Marketing and Sales');
```

- End employee current term

```sql
UPDATE dept_emp
SET to_date = CURRENT_DATE
WHERE
    dept_no IN ('d001', 'd005', 'd005', 'd008')
    AND to_date > CURRENT_DATE;
```

- Assign employees from d005, d008 → d010

```sql
INSERT INTO dept_emp (emp_no, dept_no, from_date, to_date)
SELECT
    emp_no,
    'd010',
    CURRENT_DATE,
    '9999-01-01'
FROM dept_emp
WHERE dept_no IN ('d005', 'd008') AND to_date = CURRENT_DATE;
```

- Assign employees from d001, d007 → d011

```sql
INSERT INTO dept_emp (emp_no, dept_no, from_date, to_date)
SELECT
    emp_no,
    'd011',
    CURRENT_DATE,
    '9999-01-01'
FROM dept_emp
WHERE dept_no IN ('d001', 'd007') AND to_date = CURRENT_DATE;
```

- Same for manager

```sql
UPDATE dept_manager
SET to_date = CURRENT_DATE
WHERE
    dept_no IN ('d001', 'd005', 'd005', 'd008')
    AND to_date > CURRENT_DATE;

INSERT INTO dept_manager (emp_no, dept_no, from_date, to_date)
SELECT
    emp_no,
    'd010',
    CURRENT_DATE,
    '9999-01-01'
FROM dept_emp
WHERE dept_no IN ('d005', 'd008') AND to_date = CURRENT_DATE;

INSERT INTO dept_manager (emp_no, dept_no, from_date, to_date)
SELECT
    emp_no,
    'd011',
    CURRENT_DATE,
    '9999-01-01'
FROM dept_emp
WHERE dept_no IN ('d001', 'd007') AND to_date = CURRENT_DATE;
```